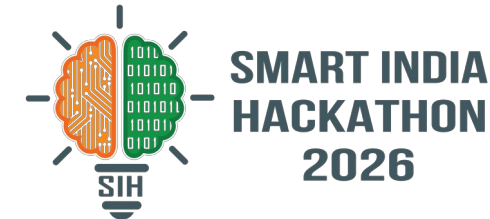


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

Problem Statement ID
MDoNER · Smart Automation · Software

PROBLEM STATEMENT ID

SIH26002

PROBLEM STATEMENT TITLE

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

THEME

Smart Automation

PS CATEGORY

Software

TEAM ID

17 (internal group no.)

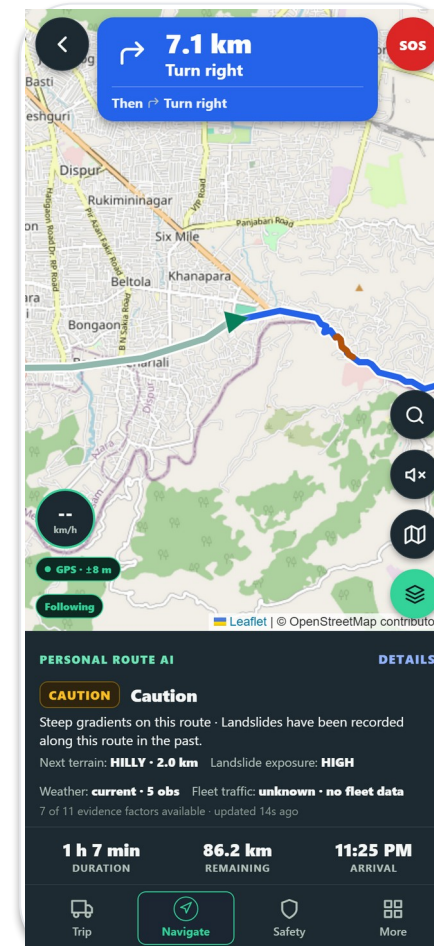
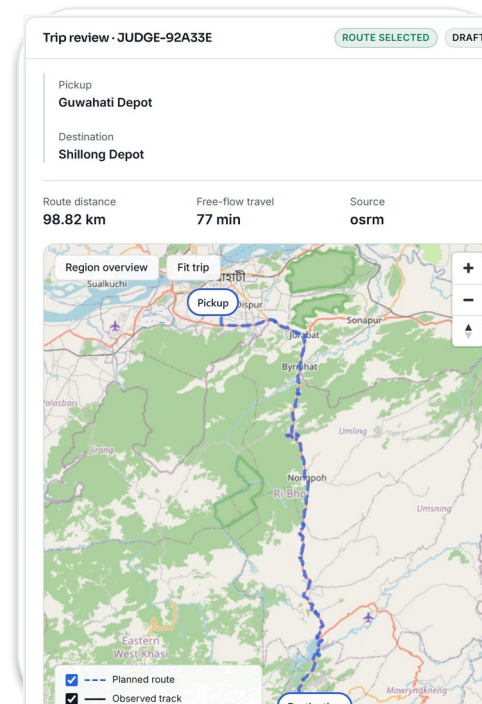
TEAM NAME

NER-AI LOGISTICS

SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong



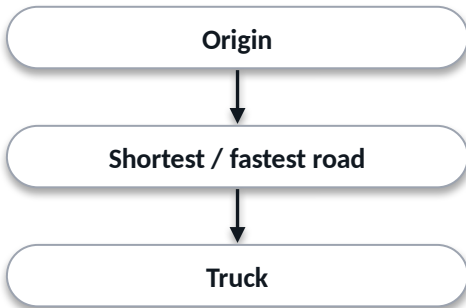
Manager route review · Driver navigation — real screens

PROPOSED SOLUTION

NOT THE SHORTEST ROAD. THE ROAD THAT IS OPERATIONALLY USABLE NOW.

NER corridors: mountain terrain · monsoon · landslides · floods · road disruption · weak connectivity · stale evidence. **A shortest-path router still sends the truck.**

A NORMAL ROUTER

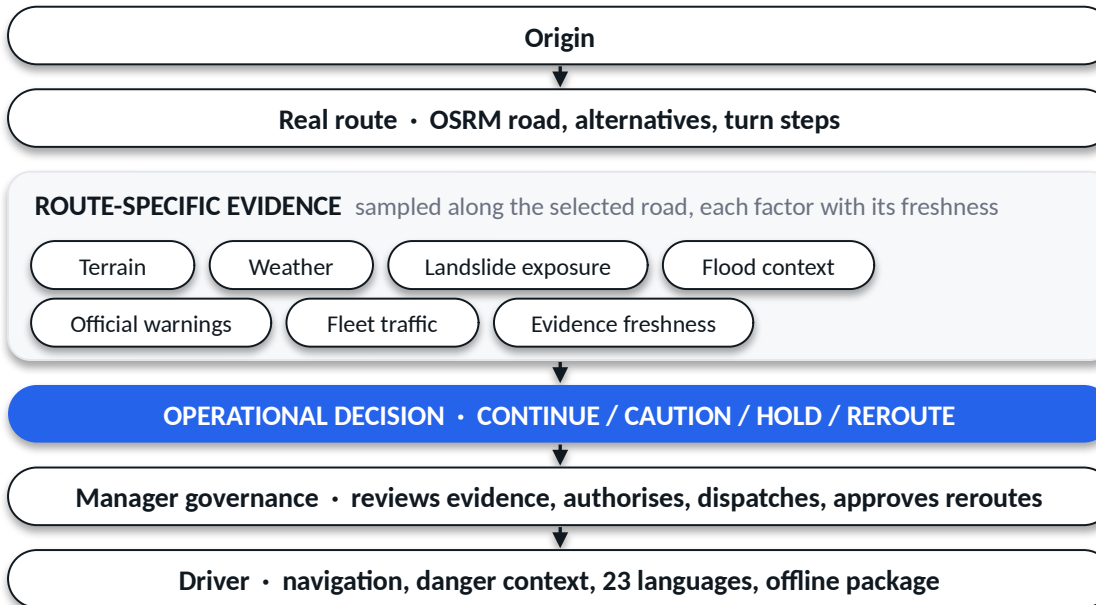


No terrain, weather, hazard or freshness check.

No governance. No answer when data is missing.

UNKNOWN ≠ SAFE

RASTA AI



The system refuses to fabricate certainty: missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0× its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

OFFICIAL ALERTS

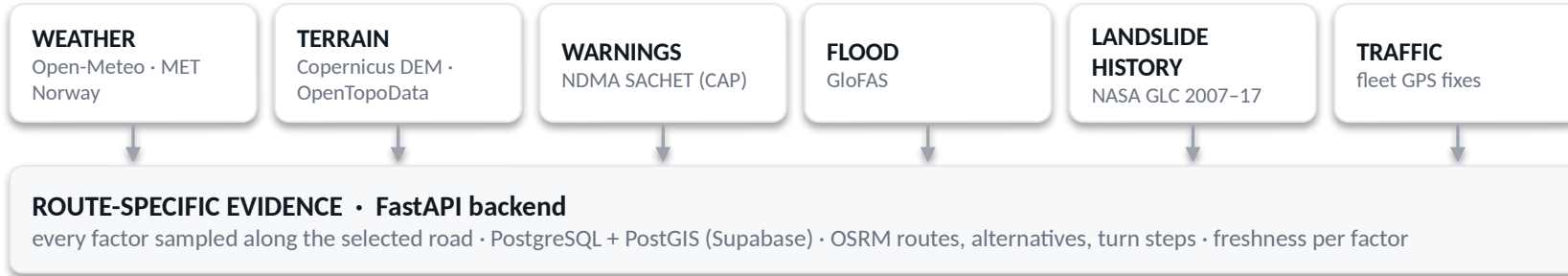
NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

Manager · Check conditions (real screen)

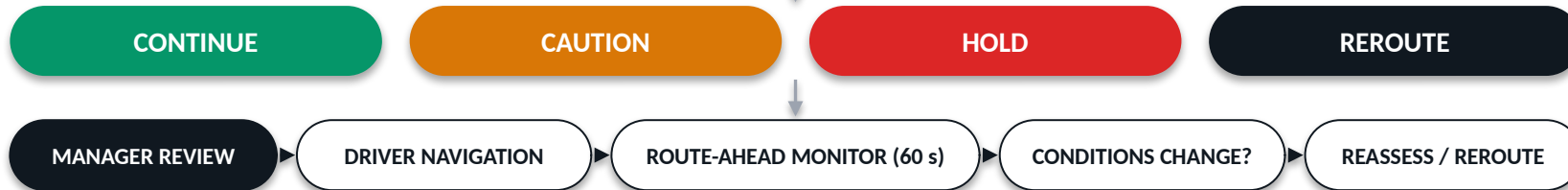
TECHNICAL APPROACH

VERIFIED DATA • provider adapters with health + freshness



DETERMINISTIC SAFETY POLICY

11 factors • reason codes • UNKNOWN ≠ SAFE • reviewer authorisation when hazard data is missing



Decision loop: conditions are re-scored every 60 s along the road ahead; a material change goes back through the policy, and a reroute is proposed to the manager — never applied silently.

CLIENTS Manager console — React + TypeScript, manager accounts only • Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package

STACK FastAPI • PostgreSQL / PostGIS (Supabase) • React • React Native / Expo • OSRM • OpenStreetMap • Gemini

EVIDENCE Open-Meteo / MET Norway • NDMA SACHET • GloFAS • NASA historical landslides • OpenTopoData / Copernicus DEM

WHO DECIDES — AI IS SUPPORTING

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision and assists in the driver's language — never decides

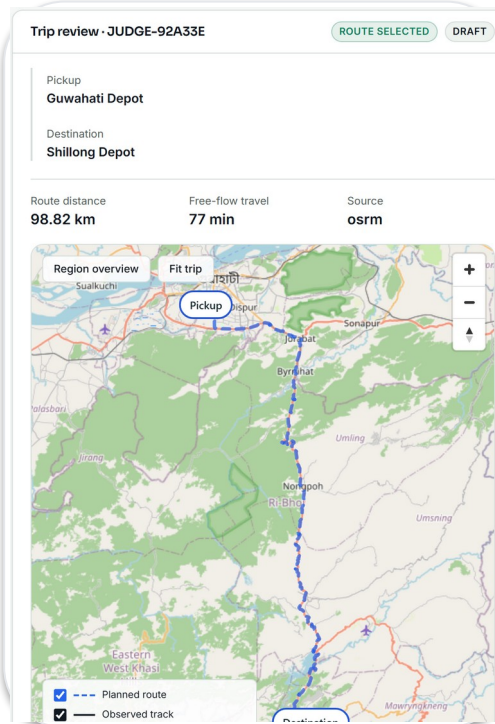
EXPERIMENTAL ML

research channel only; kept outside routing by the safety-validation gate

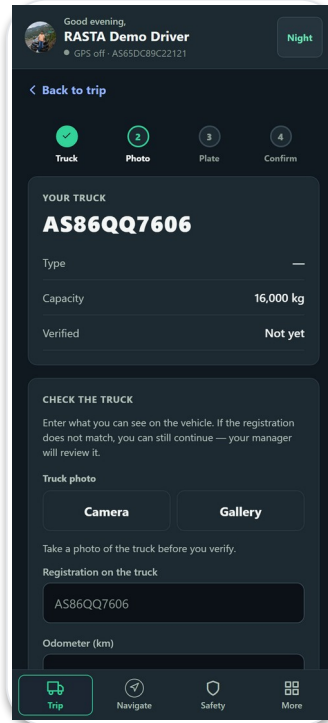
GOVERNANCE CHAIN



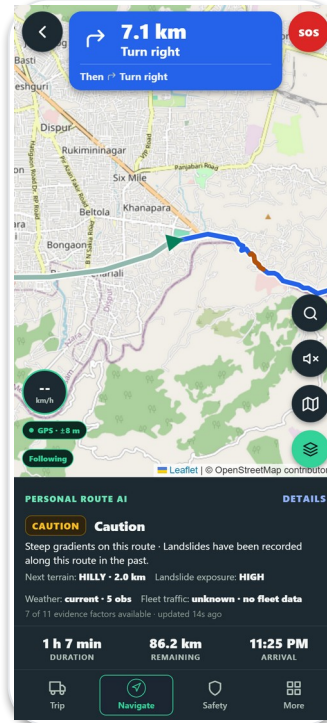
NOT A CONCEPT. A WORKING END-TO-END PROTOTYPE.



Manager · route review



Driver · truck check



Driver · navigation

PROOF

The whole lifecycle above was run on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

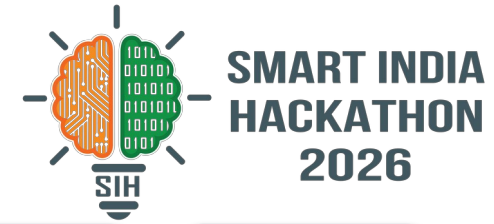
PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ~ 98.8 km real route

Every reroute on that run was approved by a manager before the phone followed it.

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

IMPACT AND BENEFITS



RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

FOOD • MEDICINE • FUEL • RELIEF SUPPLIES — essential logistics for hill communities, decided on evidence

WHO BENEFITS

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

DEMONSTRATED TODAY

23 selectable languages, explicit fallback status

11 route factors in every decision

60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

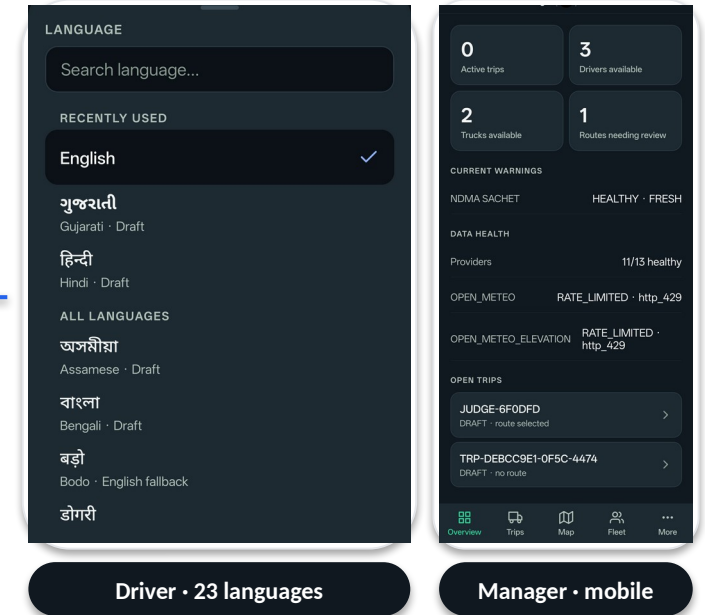
100 % of reroutes reviewed by a human

0 unknown factors counted as safe

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



No invented percentages: impact is stated as who gains what, backed by the working system.



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

OFFICIAL / FLOOD

NDMA SACHET (CAP) • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

WEATHER / TERRAIN

Open-Meteo • MET Norway • Copernicus DEM • OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

PROBLEM STATEMENT

MDoNER • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE

NOT DEPLOYED

FPR 50 % too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD_ERROR_ANALYSIS.md

REFERENCES

gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry, error analysis)